

Security Standards for Vulnerable Groups: Technical Interoperability and Blue Team Defensive Implementations

Daniel David

AI Safety & Governance Analyst
Global Multilateral AI Committee
Port Moresby, Papua New Guinea
papudaniel523@gmail.com

Lydia Thayline

AI Safety and Leader
Global Multilateral AI Committee
Initiative (Article Advisor)
Rio de Janeiro, RJ, Brazil
hello.lily.aisafety@gmail.com

Abstract—As digital threats against civil society, human rights defenders, and marginalized groups escalate in complexity, traditional perimeter-based security models have proven increasingly insufficient. This report examines the critical role of technical interoperability and proactive Blue Team defensive implementations in safeguarding these vulnerable populations. We analyze how standardized defensive frameworks—specifically protocols like STIX and TAXII—enable real-time, automated threat intelligence sharing to reduce the "digital divide" in cybersecurity capabilities. The study identifies four distinct layers of interoperability (Technical, Semantic, Organizational, and Legal) that must be aligned to establish a cohesive global shield against emerging threats like state-sponsored surveillance and targeted social engineering.

Findings suggest that interoperable standards allow for automated, cross-platform defense mechanisms that are vital for resource-constrained groups who lack traditional security infrastructure. By integrating proactive monitoring tools such as SIEM and EDR within the NIST Cybersecurity Framework, organizations can transition from passive data collection to active, real-time risk mitigation. This research highlights the necessity of Zero Trust Architecture in protecting users within politically volatile regions, ensuring data integrity regardless of network location. Ultimately, the implementation of these interoperable security standards and Blue Teaming strategies is vital for the survival of civil society in the digital age. By fostering an ecosystem where defensive tools and intelligence are shared seamlessly, we can build a collective defense that protects the most at-risk individuals from sophisticated digital adversaries.

Keywords—Blue Teaming, Technical Interoperability, Cybersecurity Standards, Vulnerability Populations, Incident Response.

I. INTRODUCTION

The rapid deployment of generative artificial intelligence Vulnerable groups, including journalists and activists, face unique digital risks ranging from state-sponsored surveillance to targeted social engineering. Unlike corporate entities, these groups often lack the infrastructure to maintain isolated security silos. Technical interoperability the ability of disparate systems to securely exchange data is therefore not just a convenience but a defensive necessity. By implementing interoperable

defensive standards, organizations can deploy "Blue Teaming" strategies that are reactive, resilient, and collaborative.

II. TECHNICAL INTEROPERABILITY STANDARDS

Interoperability allows for the automation of the sharing process, which is essential for security analysts to move from data collection to active analysis.

- A. *Information Sharing Protocols*: Standards such as STIX (Structured Threat Information eXpression) and TAXII (Trusted Automated eXchange of Intelligence Information) facilitate the automated exchange of cyber threat intelligence (CTI).
- B. *Cross-Platform Defense*: Interoperability ensures that security controls are effective across diverse environment, including cloud, IoT, and legacy systems.
- C. *Interoperability Layers*: Research has identifies four layers of interoperability; Technical, Sematic, Organization, and Legal. That must be aligned to create a global shield against emerging threats.

III. BLUE TEAMING: DEFENSIVE IMPLEMENTATION

Blue Teaming focuses on defending systems and mitigating risks through continuous monitoring and hardening.

- A. *Proactive Monitoring*: Implementing SIEM and EDR tools allows for the detection of unauthorized access in real-time, which is critical for groups facing immediate physical risks from data leaks.
- B. *Framework-Based Defense*: The NIST Cybersecurity Framework provides a five-step core (Identify, Protect, Detect, Respond, and Recover) that serves as a blueprint for blue team operations.

- C. *Zero Trust Architecture*: Moving towards Zero Trust ensures that data collection and flow are secured regardless of the network location, protecting users in politically volatile regions.

IV. PROTECTING VULNERABLE GROUPS

Security standards for marginalized populations must prioritize equity in design to avoid exacerbating existing inequalities.

- A. *Addressing Barriers*: Vulnerable groups often face limited digital literacy and systemic biases. Standards must be accessible and low-overhead.
- B. *Collaborative Protection Networks*: Transnational advocacy networks provide solidarity and self-protection capacities for human rights defenders through shared value and information exchange.
- C. *Accountability*: Standards should empower civil society to hold government and corporations accountable for digital rights violations.

V. CONCLUSION

The implementation of interoperability security standards and Blue Teaming strategies is vital for the survival of civil society in the digital age. By fostering an ecosystem where defensive tools and intelligence are shared seamlessly, we can build a collective defense that protects the most at-risk individuals from sophisticated digital adversaries.

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